KAUSTUBH KALYANRAO SURYAWANSHI

Data Science & Machine Learning Engineer | AI/ML | Python | SQL

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PROFESSIONAL SUMMARY

Data Science and Machine Learning professional with **hands-on experience** in ML analytics, demand forecasting, and computer vision. Skilled in **Python, SQL, machine learning, data analysis, and model evaluation**, with experience building and deploying practical ML solutions.

TECHNICAL SKILLS

Programming & Data: Python, SQL, Pandas, NumPy
Machine Learning: Scikit-learn, XGBoost, LightGBM, Classification, Regression, Feature Engineering, Model Evaluation,Time-Series Forecasting
AI & Analytics: NLP, AI/LLM Applications, SHAP, YOLOv8YOLO/v12, OpenCV, PyTorch, Ultralytics
APIs & Deployment: FastAPI, REST APIs, Streamlit, Docker
Databases & Cloud: PostgreSQL, MongoDB, SQLite, AWS ECS/Fargate
Visualization & Reporting: Matplotlib, Seaborn, Plotly, Excel, Power BI
Developer Tools: Git, GitHub, Jupyter Notebook

SELECTED PROJECTS

DataLens – AI Analytics Platform with Natural-Language Query Interface | GitHub

Python | *Pandas* | *Scikit-learn* | *XGBoost* | *LightGBM* | *FastAPI* | *Docker* | *AWS ECS/Fargate*

- Architected and deployed an **end-to-end ML analytics platform** with **natural-language querying**, reducing analysis time from 3+ hours to <30 seconds.
- Engineered a 4-stage pipeline for EDA, data-quality checks with PII masking, ML training on 50K+ records, and **SHAP-based explainabilit**y with What-If analysis.
- Deployed with **Docker** and **AWS ECS**/Fargate, achieving <**200ms API latency**; reduced manual analytics workload by 85**%** and supported 5**+ concurrent users**.

Walmart DemandLens – Multi-Model Demand Forecasting & Model Selection | GitHub

Python | *Pandas* | *LightGBM* | *SARIMA* | *Prophet* | *Streamlit* | *FastAPI*

- Designed a forecasting framework comparing **SARIMA, Prophet, and LightGBM** across **4 California stores** and **1K+ SKU-store combinations.**
- Engineered lag, rolling-window, calendar, promotion, price, and event features, improving **RMSSE by 15%** versus a naive baseline; selected models using validation performance.
- Built **FastAPI/Streamlit forecasting** interfaces and **28-day rolling forecasts**, reducing generation time from **2 hours to 5 minutes**; reduced stockouts **12%** and **overstock waste 8%.**

Indian Sign Language Detection & Multilingual Translation | GitHub

Python | *YOLOv8/YOLOv12* | *OpenCV* | *PyTorch* | *Ultralytics* | *PyQt5* | *Flask* | *Gemini API*

- Developed a **real-time computer-vision system** using **custom-trained YOLO models** to detect Indian Sign Language gestures from **webcam** video.
- Built **gesture buffering** and **sequence processing** to convert detected signs into text and generate coherent translations with **Gemini API**.
- Implemented **multilingual** output for **English, Hindi, Marathi, Punjabi, and Tamil** with interactive PyQt5/Flask interfaces; included **custom dataset** creation and **YOLO training workflows.**

EDUCATION

B.E. in Information Technology – University of Mumbai | 2026 | **CGPA: 7.3/10**

ACHIEVEMENTS & RECOGNITION

- **2nd Place, XZIBIT 2026 National Project Competition** — Among **150+ entries**; recognized for production deployment, scalability, and real-world applicability.
- **Finalist — 20th Aavishkar National Research Convention 2025** — Presented *SCAN-TO-KNOW (Manufactured Product Ingredient Analyzer)* in the Pure Sciences category at UG level.
- **Patent Application — SCAN-TO-KNOW: Automated Ingredient Analyzer for Packaged Foods** — Filed with the Indian Patent Office; Application No. 202621034309.

PROFESSIONAL CERTIFICATIONS

- **IBM Data Science Professional Certificate**
- **AWS Certified Cloud Practitioner**